A Finite Schur–Feshbach Test for Reflected Wilson Blocks: An Explicit Four-Dimensional $SU(2)$ Heat-Kernel Lattice Realization

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Abstract

We give a fully specified finite-cutoff realization of a Schur–Feshbach positivity test for reflected Wilson observables. The matrix statement is the rank-one specialization of standard second-order geometry of the positive-semidefinite cone: if

$$q_\varepsilon = \mathbf{1}\mathbf{1}^* + \varepsilon H + \varepsilon^2 K + o(\varepsilon^2),$$

then, after eliminating the rank-one direction, the first two coefficients are

$$A = PHP, \qquad C = P\big(K - n^{-1} H e e^* H\big)P, \qquad e = n^{-1/2}\mathbf{1}, \quad P = I - ee^*.$$

The field-theoretic contribution is an explicit same-theory block. On a finite four-dimensional reflection-symmetric lattice with gauge group $SU(2)$ and heat-kernel plaquette weight, we define three closed hairpin loops sharing a base path. A spatial-reflection odd Wilson mode X and an even centered Wilson mode Y generate three positive-time observables

$$F_i^\varepsilon = 1 + \sqrt{\varepsilon}\, b_i X + \varepsilon c_i Y, \qquad b = (1,-1,0)^T, \quad c = (0,0,1)^T.$$

Their reflected Gram matrix is exactly

$$N_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon\sigma b b^T + \varepsilon^2 \tau c c^T, \qquad \sigma = \langle \Theta X, X\rangle_{\mathrm{OS}} \geq 0, \quad \tau = \langle \Theta Y, Y\rangle_{\mathrm{OS}} \geq 0.$$

Diagonal correlation normalization is realized by rescaling the same observables, hence is an exact congruence rather than an uncontrolled entrywise quotient. The normalized block has a rank-one limit and yields

$$PHP = \sigma b b^T, \qquad C = -\sigma^2 b b^T + \frac{\tau}{9} z z^T, \qquad z = (1,1,-2)^T.$$

Thus the first reduced eigenvalue is 2σ, while the corrected second-order null-sector eigenvalue is $2\tau/3$. The uncorrected quantity contains a spurious σ^2 term, which the Schur–Feshbach correction removes. All statements up to this point are exact at finite cutoff. Geometric collar collapse, uniform nondegeneracy along a continuum trajectory, endpoint cumulant estimates, infinite volume, and a spectral-gap implication are isolated as explicit open modules rather than claimed results.

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1 Scope, novelty, and claim boundary

The Yang–Mills existence and mass-gap problem asks for a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group and a positive mass gap in the same reconstructed theory [10, 11]. Osterwalder–Schrader reconstruction provides a Euclidean route, in which reflection positivity is one indispensable axiom [2, 3]. At finite lattice spacing, standard plaquette gauge actions furnish reflection-positive regulated theories [6, 5].

This paper makes a deliberately narrower contribution. It does not construct a continuum Yang–Mills measure, establish all OS axioms, prove exponential clustering, or prove a mass gap. It does two finite things:

(i) it states the rank-one Schur–Feshbach coefficient test for a singular PSD matrix family; and

(ii) it implements that test on an explicit four-dimensional $SU(2)$ lattice Wilson block whose action, reflection, observables, normalization, rank-one limit, coefficients, and finite margins are all specified.

The Schur-complement curvature term is not presented as a new discovery in matrix analysis. It is a rank-one instance of the established second-order geometry of semidefinite constraints [8, 9]. The purpose of retaining the finite theorem is to make the Wilson calculation self-contained and to show exactly where the curvature correction enters. The substantive advance over a purely schematic Wilson/OS discussion is the same-theory realization in Sections 3–5.

Claim boundary. All results through Corollary 5.4 hold on a fixed finite lattice, and the source-collision expansion is uniform over arbitrary cutoff families because the primitive Wilson observables are bounded. Strict positive margins require the finite nondegeneracy conditions $\sigma > 0$ and $\tau > 0$. No uniform lower bound on these quantities as the cutoff is removed is proved here.

2 Finite reflected Gram matrices and the Schur–Feshbach test

Let Θ be an antilinear Euclidean reflection and let $\mathcal{A}_+$ be an algebra of positive-time observables. Reflection positivity means

$$\sum_{i,j=1}^{n} \overline{c_i} c_j \, \langle (\Theta F_i)^* F_j \rangle \geq 0 \tag{1}$$

for every finite family $F_1, \ldots, F_n \in \mathcal{A}_+$ and every $c \in \mathbb{C}^n$. Equivalently, the reflected Gram matrix

$$N_{ij} = \langle (\Theta F_i)^* F_j \rangle \tag{2}$$

is positive semidefinite.

Suppose a normalized finite block has a rank-one collision expansion

$$q_\varepsilon = \mathbf{1}\mathbf{1}^* + \varepsilon H + \varepsilon^2 K + R_\varepsilon, \qquad \|R_\varepsilon\| = o(\varepsilon^2), \tag{3}$$

where $H, K \in \mathrm{Herm}_n(\mathbb{C})$. Define

$$e = n^{-1/2}\mathbf{1}, \qquad P = I - ee^*. \tag{4}$$

Then P projects onto $\mathbf{1}^\perp$ and $\mathbf{1}\mathbf{1}^* = nee^*$.

Theorem 2.1 (Rank-one Schur–Feshbach certificate)**.** *Let q_ε satisfy (3). Set*

$$\lambda_\varepsilon = e^* q_\varepsilon e, \qquad \mathcal{F}_\varepsilon = P q_\varepsilon P - P q_\varepsilon e \, \lambda_\varepsilon^{-1} e^* q_\varepsilon P. \tag{5}$$

For all sufficiently small $\varepsilon > 0$, $\lambda_\varepsilon > 0$ and

$$q_\varepsilon \succeq 0 \quad \Longleftrightarrow \quad \mathcal{F}_\varepsilon \succeq 0. \tag{6}$$

Moreover,

$$\mathcal{F}_\varepsilon = \varepsilon A + \varepsilon^2 C + o(\varepsilon^2), \qquad A = PHP, \qquad C = P\big(K - n^{-1} H e e^* H\big)P. \tag{7}$$

If $q_\varepsilon \succeq 0$ for all sufficiently small $\varepsilon > 0$, then

$$A \succeq 0 \quad \text{on } \mathbf{1}^\perp, \tag{8}$$

and, writing Π for the orthogonal projection onto $\operatorname{Ker} A \subset \mathbf{1}^\perp$,

$$\Pi C \Pi \succeq 0 \quad \text{on } \operatorname{Ker} A. \tag{9}$$

Proof. Since $q_0 = nee^*$,

$$\lambda_\varepsilon = n + \varepsilon e^* H e + O(\varepsilon^2) > 0$$

for small ε. With respect to $\mathbb{C}^n = \mathbb{C}e \oplus P\mathbb{C}^n$, the usual scalar Schur complement gives (6). Furthermore,

$$\begin{aligned} Pq_\varepsilon P &= \varepsilon PHP + \varepsilon^2 PKP + o(\varepsilon^2), \\ Pq_\varepsilon e &= \varepsilon PHe + \varepsilon^2 PKe + o(\varepsilon^2), \\ \lambda_\varepsilon^{-1} &= n^{-1} + O(\varepsilon). \end{aligned}$$

Hence

$$Pq_\varepsilon e\, \lambda_\varepsilon^{-1} e^* q_\varepsilon P = \varepsilon^2 n^{-1} PHee^*HP + o(\varepsilon^2),$$

which proves (7). If $\mathcal{F}_\varepsilon \succeq 0$, then for $v \in \mathbf{1}^\perp$,

$$0 \le v^* \mathcal{F}_\varepsilon v = \varepsilon v^* A v + \varepsilon^2 v^* C v + o(\varepsilon^2).$$

Division first by ε, and then by ε^2 on $\operatorname{Ker} A$, gives (8) and (9). □

Remark 2.2 (The curvature correction)**.** *Since* $ee^* = n^{-1}\mathbf{1}\mathbf{1}^*$,

$$C = PKP - n^{-2} PH\mathbf{1}\mathbf{1}^* HP. \tag{10}$$

The second term is the excursion from $\mathbf{1}^\perp$ *into the rank-one direction and back. Replacing* C *by* PKP *generally changes the null-sector value.*

Proposition 2.3 (Strict second-order sufficiency)**.** *Assume* $A \succeq 0$ *on* $\mathbf{1}^\perp$*. Let* Π *project onto* $\operatorname{Ker} A$ *and put* $Q = P - \Pi$*. If, for some* $\alpha, \beta > 0$*,*

$$QAQ \succeq \alpha Q, \qquad \Pi C \Pi \succeq \beta \Pi, \tag{11}$$

then $q_\varepsilon \succeq 0$ *for all sufficiently small* $\varepsilon > 0$*.*

Proof. On $Q\mathbb{C}^n$, $Q\mathcal{F}_\varepsilon Q = \varepsilon QAQ + O(\varepsilon^2)$ is positive and has inverse of order $O(\varepsilon^{-1})$. Because $A\Pi = 0 = \Pi A$, the off-diagonal block $\Pi \mathcal{F}_\varepsilon Q$ is $O(\varepsilon^2)$. The Schur complement on $\Pi\mathbb{C}^n$ is therefore

$$\varepsilon^2 \Pi C \Pi + o(\varepsilon^2) - O(\varepsilon^3),$$

which is positive for small ε by (11). Theorem 2.1 then gives $q_\varepsilon \succeq 0$. □

Remark 2.4 (Higher-order recursion requires higher-order data)**.** *The two-order hypothesis* (3) *supports only the two tests above. A recursive null-sector procedure is justified only when an asymptotic expansion through the required order, or analyticity in* ε*, is available. An* $o(\varepsilon^2)$ *remainder alone does not determine the next sign.*

3 A reflection-positive four-dimensional lattice model

3.1 Finite lattice, action, and reflections

Fix a lattice spacing $a > 0$, integers $T, L \ge 1$, and the finite reflection-symmetric lattice

$$\Lambda_{T,L} = \{-T+1, \dots, T\} \times (\mathbb{Z}/L\mathbb{Z})^3. \tag{12}$$

The first coordinate is Euclidean time. An oriented nearest-neighbour edge ℓ carries a link variable $U_\ell \in SU(2)$, with $U_{\bar{\ell}} = U_\ell^{-1}$. For a plaquette p, U_p is the ordered product of link variables around its boundary.

Let K_κ be the $SU(2)$ heat kernel at time $\kappa > 0$, in the convention

$$K_\kappa(g) = \sum_{j \in \frac{1}{2}\mathbb{N}_0} (2j+1) e^{-\kappa j(j+1)} \chi_j(g). \tag{13}$$

The finite-volume heat-kernel plaquette measure is

$$d\mu_{\Lambda,\kappa}(U) = Z_{\Lambda,\kappa}^{-1} \left(\prod_\ell dU_\ell \right) \left(\prod_p K_\kappa(U_p) \right), \tag{14}$$

where dU_ℓ is normalized Haar measure. This is a finite-dimensional probability measure. The heat-kernel action is a standard lattice Yang–Mills action and has strictly positive character coefficients [7].

Reflect the lattice through the link plane $t = \frac{1}{2}$:

$$\vartheta(t, x_1, x_2, x_3) = (1 - t, x_1, x_2, x_3). \tag{15}$$

The induced antilinear operation on observables is denoted by Θ. Let $\mathcal{A}_+$ be the gauge-invariant Wilson algebra generated by loops whose links lie entirely in the positive half $t \geq 1$.

Proposition 3.1 (Finite reflection positivity)**.** *For the measure* (14)*,*

$$\langle (\Theta F)^* F \rangle_{\Lambda,\kappa} \geq 0 \qquad (F \in \mathcal{A}_+). \tag{16}$$

Consequently every finite reflected Wilson Gram matrix is positive semidefinite.

Justification. This is the standard link-reflection positivity theorem for plaquette gauge actions with nonnegative character coefficients [6, 5]. In the heat-kernel case the coefficients in (13) are positive. The Peter–Weyl expansion factorizes each plaquette crossing the reflection plane into a sum of reflected products with nonnegative weights; plaquettes entirely on one side occur in reflected pairs. Expanding the finite product and integrating gives (16). □

We also use the spatial reflection

$$\rho(t, x_1, x_2, x_3) = (t, x_1, -x_2, x_3). \tag{17}$$

The measure is invariant under ρ, and ρ commutes with ϑ.

3.2 Three explicit closed hairpins

Let e_0, e_1, e_2, e_3 be the lattice unit vectors, with e_0 temporal. Choose positive integers r, s, h so that the following paths do not wrap around the finite box. Put

$$A = (1, 0, 0, 0), \qquad B = A + re_1,$$

and let β be the straight path from A to B. Define three detours from A to B:

$$\eta_+ : A \longrightarrow A + se_2 \longrightarrow B + se_2 \longrightarrow B, \tag{18}$$

$$\eta_- : A \longrightarrow A - se_2 \longrightarrow B - se_2 \longrightarrow B, \tag{19}$$

$$\eta_0 : A \longrightarrow A + he_0 \longrightarrow B + he_0 \longrightarrow B. \tag{20}$$

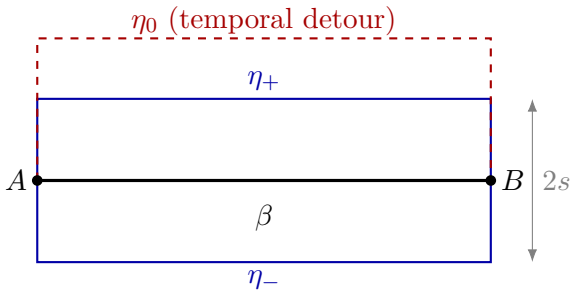


Figure 1: Schematic projection of the three detours. The dashed detour η_0 lies in the Euclidean-time direction; the picture only displays the shared-base incidence pattern.

The closed hairpin loops are

$$\gamma_\alpha = \eta_\alpha \circ \beta^{-1}, \qquad \alpha \in \{+, -, 0\}. \tag{21}$$

All links lie in $t \geq 1$. The loops share the base path β, and

$$\rho\gamma_+ = \gamma_-, \qquad \rho\gamma_0 = \gamma_0. \tag{22}$$

For a closed path γ, define the normalized fundamental Wilson loop

$$W_\gamma(U) = \frac{1}{2}\operatorname{Tr} U_\gamma. \tag{23}$$

For $SU(2)$ this is real and lies in $[-1, 1]$. Write $W_+ = W_{\gamma_+}$, $W_- = W_{\gamma_-}$, and $W_0 = W_{\gamma_0}$. Define

$$X = \frac{1}{2}(W_+ - W_-), \qquad Y = W_0 - m_0, \qquad m_0 = \langle W_0 \rangle_{\Lambda,\kappa}. \tag{24}$$

Then $X, Y \in \mathcal{A}_+$, $|X| \leq 1$, $|Y| \leq 2$, and

$$\rho X = -X, \qquad \rho Y = Y, \qquad \langle X \rangle = \langle Y \rangle = 0. \tag{25}$$

Lemma 3.2 (Symmetry decoupling)**.** *The mixed reflected correlations vanish:*

$$\langle (\Theta X)Y \rangle_{\Lambda,\kappa} = 0, \qquad \langle (\Theta Y)X \rangle_{\Lambda,\kappa} = 0. \tag{26}$$

Proof. Because the measure is ρ-invariant and ρ commutes with Θ,

$$\langle (\Theta X)Y \rangle = \langle (\Theta\rho X)(\rho Y) \rangle = -\langle (\Theta X)Y \rangle.$$

Thus the first correlation is zero. The second is its Hermitian conjugate and also vanishes. □

4 The exact three-hairpin source block

4.1 Positive-time observables and exact Gram matrix

Let

$$b = (1, -1, 0)^T, \qquad c = (0, 0, 1)^T, \qquad z = (1, 1, -2)^T. \tag{27}$$

Thus $b \perp \mathbf{1}$, $z \perp \mathbf{1}$, and $b \perp z$. For $\varepsilon \geq 0$, define three observables in the same positive-time Wilson algebra:

$$F_i^\varepsilon = 1 + \sqrt{\varepsilon}\, b_i X + \varepsilon c_i Y, \qquad i = 1, 2, 3. \tag{28}$$

The square root appears only at the observable level. The ρ parity in Lemma 3.2 removes half-integer powers from the Gram matrix.

Set

$$\sigma = \langle (\Theta X) X \rangle_{\Lambda,\kappa}, \qquad \tau = \langle (\Theta Y) Y \rangle_{\Lambda,\kappa}. \tag{29}$$

By Proposition 3.1,

$$0 \leq \sigma \leq 1, \qquad 0 \leq \tau \leq 4. \tag{30}$$

The upper bounds follow from $|X| \leq 1$ and $|Y| \leq 2$.

Proposition 4.1 (Exact reflected Gram block)**.** *The reflected Gram matrix*

$$N_\varepsilon(i,j) = \left\langle (\Theta F_i^\varepsilon)^* F_j^\varepsilon \right\rangle_{\Lambda,\kappa} \tag{31}$$

is exactly

$$N_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon\sigma b b^T + \varepsilon^2 \tau c c^T. \tag{32}$$

In particular, $N_\varepsilon \succeq 0$ for every $\varepsilon \geq 0$.

Proof. Expanding (31), the one-point terms vanish by (25). The order-$\varepsilon^{3/2}$ mixed terms vanish by (26). The remaining terms are precisely those in (32); there are no higher terms because (28) is polynomial of degree one in $\sqrt{\varepsilon}X$ and degree one in εY. Positivity also follows directly from

$$v^T N_\varepsilon v = |\mathbf{1}^T v|^2 + \varepsilon\sigma |b^T v|^2 + \varepsilon^2 \tau |c^T v|^2.$$

□

4.2 Same-theory correlation normalization

The diagonal entries of (32) are

$$N_{11} = N_{22} = 1 + \varepsilon\sigma, \qquad N_{33} = 1 + \varepsilon^2\tau. \tag{33}$$

Define

$$D_\varepsilon = \operatorname{diag}\left((1+\varepsilon\sigma)^{-1/2}, (1+\varepsilon\sigma)^{-1/2}, (1+\varepsilon^2\tau)^{-1/2} \right) \tag{34}$$

and normalized observables

$$\widetilde{F}_i^\varepsilon = (D_\varepsilon)_{ii} F_i^\varepsilon. \tag{35}$$

Their reflected Gram matrix is

$$q_\varepsilon = D_\varepsilon N_\varepsilon D_\varepsilon, \qquad (q_\varepsilon)_{ij} = \frac{N_\varepsilon(i,j)}{\sqrt{N_\varepsilon(i,i) N_\varepsilon(j,j)}}. \tag{36}$$

This is an exact observable-level congruence. It is therefore positivity preserving and does not invoke an arbitrary pair-dependent quotient.

Proposition 4.2 (Exact normalized block)**.** *The normalized matrix is*

$$q_\varepsilon = \begin{pmatrix} 1 & \alpha_\varepsilon & \beta_\varepsilon \\ \alpha_\varepsilon & 1 & \beta_\varepsilon \\ \beta_\varepsilon & \beta_\varepsilon & 1 \end{pmatrix}, \tag{37}$$

where

$$\alpha_\varepsilon = \frac{1-\varepsilon\sigma}{1+\varepsilon\sigma}, \qquad \beta_\varepsilon = \frac{1}{\sqrt{(1+\varepsilon\sigma)(1+\varepsilon^2\tau)}}. \tag{38}$$

It satisfies $q_\varepsilon \succeq 0$ for every $\varepsilon \geq 0$ and

$$q_\varepsilon \longrightarrow \mathbf{1}\mathbf{1}^T \qquad (\varepsilon \downarrow 0). \tag{39}$$

If $\sigma > 0$ and $\tau > 0$, then $q_\varepsilon \succ 0$ for every $\varepsilon > 0$, and

$$\det q_\varepsilon = \frac{4\varepsilon^3\sigma\tau}{(1+\varepsilon\sigma)^2(1+\varepsilon^2\tau)}. \tag{40}$$

Proof. Equations (37)–(38) follow from (32) and (34). Positivity and the rank-one limit follow from the congruence (36). If $\sigma, \tau > 0$ and $\varepsilon > 0$, the three linear functionals $\mathbf{1}^T v$, $b^T v$, and $c^T v$ separate points of $\mathbb{R}^3$, so (32) is positive definite. Direct calculation gives (40). □

5 Coefficient extraction and the finite certificate

Expanding (38) at $\varepsilon = 0$ gives

$$\alpha_\varepsilon = 1 - 2\sigma\varepsilon + 2\sigma^2\varepsilon^2 + O(\varepsilon^3), \tag{41}$$

$$\beta_\varepsilon = 1 - \frac{\sigma}{2}\varepsilon + \left(\frac{3\sigma^2}{8} - \frac{\tau}{2}\right)\varepsilon^2 + O(\varepsilon^3). \tag{42}$$

Thus

$$q_\varepsilon = \mathbf{1}\mathbf{1}^T + \varepsilon H + \varepsilon^2 K + R_\varepsilon, \tag{43}$$

with

$$H = \begin{pmatrix} 0 & -2\sigma & -\sigma/2 \\ -2\sigma & 0 & -\sigma/2 \\ -\sigma/2 & -\sigma/2 & 0 \end{pmatrix} \tag{44}$$

and

$$K = \begin{pmatrix} 0 & 2\sigma^2 & 3\sigma^2/8 - \tau/2 \\ 2\sigma^2 & 0 & 3\sigma^2/8 - \tau/2 \\ 3\sigma^2/8 - \tau/2 & 3\sigma^2/8 - \tau/2 & 0 \end{pmatrix}. \tag{45}$$

Proposition 5.1 (Uniform source-collision remainder)**.** *There is a numerical constant $C_* > 0$, independent of a, T, L, κ, r, s, h, such that*

$$\|R_\varepsilon\|_{\mathrm{op}} \leq C_*\varepsilon^3 \qquad (0 \leq \varepsilon \leq 1/4). \tag{46}$$

Proof. By (30), (σ, τ) lies in the compact set $[0,1] \times [0,4]$. The functions in (38) and their first three ε derivatives are continuous on

$$[0, 1/4] \times [0,1] \times [0,4].$$

Taylor's theorem therefore gives a uniform cubic entrywise remainder. A 3×3 Hermitian matrix has operator norm bounded by its maximum absolute row sum, which converts the entrywise bound into (46). Appendix B records an explicit admissible definition of C_*. □

Let

$$e = 3^{-1/2}\mathbf{1}, \qquad P = I - ee^T, \qquad u = 2^{-1/2}b, \qquad v = 6^{-1/2}z. \tag{47}$$

Then u, v form an orthonormal basis of $\mathbf{1}^\perp$.

Theorem 5.2 (Evaluated Schur–Feshbach gate)**.** *For the explicit block* (37)*, the first reduced coefficient is*

$$A = PHP = \sigma bb^T = 2\sigma uu^T. \tag{48}$$

The corrected second coefficient is

$$C = P\left(K - \frac{1}{3}Hee^TH\right)P = -\sigma^2 bb^T + \frac{\tau}{9}zz^T = -2\sigma^2 uu^T + \frac{2\tau}{3}vv^T. \tag{49}$$

Consequently:

(a) $A \succeq 0$ *on* $\mathbf{1}^\perp$*;*

(b) if $\sigma > 0$*, then* $\operatorname{Ker} A = \operatorname{span}\{z\}$ *on* $\mathbf{1}^\perp$*;*

(c) on that null sector,

$$z^TCz = 4\tau, \qquad v^TCv = \frac{2\tau}{3}; \tag{50}$$

(d) if $\sigma > 0$ *and* $\tau > 0$*, Proposition 2.3 applies with*

$$\alpha = 2\sigma, \qquad \beta = \frac{2\tau}{3}. \tag{51}$$

Proof. Because $b \perp \mathbf{1}$, the additive one-body terms introduced by diagonal normalization vanish after projection, and direct multiplication gives (48). Substituting (44) and (45) into Theorem 2.1 gives (49). Since $b \perp z$, the stated eigenvalues and null-sector values follow immediately. □

Remark 5.3 (The correction removes a concrete spurious term)**.** *For the null vector* z*,*

$$z^TPKPz = \sigma^2 + 4\tau, \tag{52}$$

$$z^T\left(\frac{1}{3}PHee^THP\right)z = \sigma^2, \tag{53}$$

$$z^TCz = 4\tau. \tag{54}$$

Thus the vacuum-direction coupling contributes exactly the σ^2 *term. In this example the uncorrected sign happens to remain nonnegative, but its value is wrong and would give an incorrect quantitative margin.*

Corollary 5.4 (Implemented finite Route A)**.** *For every fixed finite lattice* (12)*, every heat-kernel parameter* $\kappa > 0$*, and every admissible hairpin geometry* (18)–(21)*, the following items are established within one and the same regulated* $SU(2)$ *gauge theory:*

(i) an explicit gauge-invariant positive-time Wilson family;

(ii) the standard OS reflection used by the lattice theory;

(iii) an exact reflected Gram matrix;

(iv) a positivity-preserving normalization realized by rescaling those observables;

(v) an exact rank-one collision limit;

(vi) explicit H and K with a cutoff-uniform operator-norm remainder;

(vii) an evaluated first gate 2σ and corrected null-sector gate $2\tau/3$.

No proxy measure or pair-dependent counterterm is introduced.

6 Finite evaluation and reproducibility protocol

The only model-dependent numbers entering the closed-form block are the finite Haar integrals

$$\sigma = Z_{\Lambda,\kappa}^{-1} \int (\Theta X)(U) X(U) \prod_{\ell} dU_\ell \prod_{p} K_\kappa(U_p), \tag{55}$$

$$\tau = Z_{\Lambda,\kappa}^{-1} \int (\Theta Y)(U) Y(U) \prod_{\ell} dU_\ell \prod_{p} K_\kappa(U_p). \tag{56}$$

They can be evaluated by a finite character expansion, exact integration on a sufficiently small lattice, or reflection-symmetric Monte Carlo. Nonnegativity is already rigorous; numerical work is needed only to quantify strict margins at selected parameters.

A minimal computational record should report

$$(a, T, L, \kappa, r, s, h), \qquad \widehat{\sigma} \pm \delta_\sigma, \qquad \widehat{\tau} \pm \delta_\tau,$$

and should preserve the reflected pairing when estimating (55)–(56). A certified strict finite margin follows from

$$\widehat{\sigma} - \delta_\sigma > 0, \qquad \widehat{\tau} - \delta_\tau > 0. \tag{57}$$

The result-elaboration certificate `routea_three_hairpin_certificate.json`, generated from the supplied verification script, symbolically verifies the formulas for N_ε, q_ε, H, K, A, C, the determinant, the uncorrected null value $z^T PKPz = \sigma^2 + 4\tau$, the removed Schur–Feshbach term σ^2, and the corrected margins $u^T Au = 2\sigma$ and $v^T Cv = 2\tau/3$. This certificate supports the finite source-collision Route A block only; it is not evidence for the geometric collar modules G1–G4, endpoint modules E1–E2, or continuum/gap modules M1–M2.

7 Open modules toward a geometric collar and a mass-gap programme

The source parameter ε in (28) makes the finite block collide while the three primitive loops remain geometrically fixed. This is sufficient to instantiate the matrix theorem and validate the same-theory normalization, but it is not yet the geometric collar collapse contemplated in the original programme. The remaining steps are stated here in a form that can be filled without changing the finite result.

7.1 Cutoff trajectory and geometric collision

A continuum-facing family would require sequences

$$a_k \downarrow 0, \qquad L_k a_k \longrightarrow \infty, \qquad T_k a_k \longrightarrow \infty, \qquad \kappa_k = \kappa(a_k), \tag{58}$$

Requirement	Status in this paper	Location
Gauge group and regulator	$SU(2)$ heat-kernel plaquette measure on a finite four-dimensional lattice	Section 3
Reflection map and positive algebra	Link reflection $t \mapsto 1-t$ and Wilson algebra supported in $t \geq 1$	Proposition 3.1
Explicit Wilson geometry	Three closed hairpins sharing β	Section 3.2
Same-theory normalization	Normalized observables $\widetilde{F}_i^{\varepsilon}$; exact congruence	Equation (36)
Coefficient extraction	Closed-form H, K, and uniform $O(\varepsilon^3)$ remainder	Equations (44)–(46)
Finite test	$A = 2\sigma uu^T$, $C = -2\sigma^2 uu^T + (2\tau/3)vv^T$	Theorem 5.2
Strict finite margins	Conditional only on finite nondegeneracy $\sigma, \tau > 0$	Equation (51)
Continuum and mass gap	Not proved; separated into open modules	Section 7

Table 1: Implemented and open components of the concrete Route A realization.

with integer hairpin dimensions scaled to prescribed physical lengths, for example

$$a_k r_k \to R, \qquad a_k s_k \to S, \qquad a_k h_k \to H. \tag{59}$$

A genuine collar parameter δ would then enter through paths $\gamma_{i,k}(\delta)$ rather than through source amplitudes alone.

Open module G1: geometric family. Specify lattice paths $\gamma_{i,k}(\delta)$ that share a locked boundary footprint, remain in the positive algebra, and converge to the intended continuum Wilson geometry as $a_k \to 0$ and $\delta \downarrow 0$. The discretization must state the order or joint scaling of a_k and δ.

Open module G2: renormalized same-theory normalization. Show that perimeter, cusp, and boundary factors are removed by observable rescaling or another positivity-preserving factorization. A pair-dependent entrywise quotient is not admissible without a separate positive-kernel theorem.

Open module G3: geometric coefficient extraction. Derive, in operator norm for a fixed finite block,

$$q_{k,\delta} = \mathbf{1}\mathbf{1}^T + \delta H_k + \delta^2 K_k + R_{k,\delta}, \qquad \sup_k \frac{\|R_{k,\delta}\|}{\delta^2} \longrightarrow 0.$$

The source-collision family already has a cutoff-uniform cubic remainder, but this does not establish the corresponding geometric expansion.

Open module G4: uniform nondegeneracy. Determine the cutoff scaling of σ_k and τ_k, including any Wilson-loop renormalization, and prove the lower bounds needed for a nonvanishing reduced margin. Boundedness gives uniform upper bounds, not the required lower bounds.

7.2 Correct endpoint/null-channel bookkeeping

Let R be a real symmetric coefficient matrix in a normalized 3×3 block and let $z = (1, 1, -2)^T$. The exact identity is

$$z^T R z = R_{11} + R_{22} + 4R_{33} + 2R_{12} - 4R_{13} - 4R_{23}. \tag{60}$$

Only when the normalization enforces $R_{11} = R_{22} = R_{33} = 0$ does this reduce to the incidence weights

$$z^T R z = 2R_{12} - 4R_{13} - 4R_{23}. \tag{61}$$

The exact block (37) has unit diagonal for every ε, so every coefficient beyond the leading term has zero diagonal and (61) applies.

Suppose an endpoint calculation produces a computable approximation $\widehat{R}_{\mathrm{end}}$ and a certified two-sided error

$$|R_{\mathrm{end}} - \widehat{R}_{\mathrm{end}}| \le \Delta_{\mathrm{end}}. \tag{62}$$

Lemma 7.1 (Logically valid endpoint certificates)**.** *Let M_G be the finite positive margin before the endpoint correction. Under (62),*

$$M_G + \widehat{R}_{\mathrm{end}} - \Delta_{\mathrm{end}} > 0 \quad \Longrightarrow \quad M_G + R_{\mathrm{end}} > 0, \tag{63}$$

$$M_G + \widehat{R}_{\mathrm{end}} + \Delta_{\mathrm{end}} < 0 \quad \Longrightarrow \quad M_G + R_{\mathrm{end}} < 0. \tag{64}$$

If one has only a lower bound $R_{\mathrm{end}} \ge -U$, then $M_G - U > 0$ is a sufficient positive certificate, while $U > M_G$ is inconclusive.

Proof. Equation (62) gives

$$\widehat{R}_{\mathrm{end}} - \Delta_{\mathrm{end}} \le R_{\mathrm{end}} \le \widehat{R}_{\mathrm{end}} + \Delta_{\mathrm{end}}.$$

Adding M_G proves both implications. A one-sided lower bound cannot certify negativity. □

A heat-kernel endpoint implementation could seek an estimator of the schematic form

$$\widehat{R}_{\mathrm{end}} = C_{\mathrm{col}} \sum_v \left(2\widehat{\kappa}^{12}_{4,v} - 4\widehat{\kappa}^{13}_{4,v} - 4\widehat{\kappa}^{23}_{4,v} \right) + \widehat{E}_{\mathrm{tail}}, \tag{65}$$

with a two-sided error obtained from source-derivative and tail estimates.

> **Open module E1: endpoint regulator derivation.** Define the endpoint boxes, source variables, singlet projections, color factor C_{col}, and tail region directly from the geometric Wilson family in Open module G1. Derive (65); do not assume it as an incidence ansatz.

> **Open module E2: cumulant and tail bounds.** Bound the connected fourth source derivatives and all omitted endpoint, corner, and inter-box terms on both sides. The five-point formula in Appendix C controls discretization of a fourth derivative once a uniform sixth-derivative bound is available; obtaining that bound is part of this module.

7.3 From OS positivity to a spectral gap

Reflection positivity of finite blocks is necessary for OS reconstruction but does not imply a spectral gap. A mass-gap argument must additionally produce the continuum Schwinger functions, the reconstructed Hilbert space and Hamiltonian, and a uniform decay or spectral estimate in that same theory.

> **Open module M1: continuum reconstruction.** Prove existence of an infinite-volume continuum limit satisfying the full OS axioms for a separating Wilson algebra, not only positivity of the finite block considered here.

> **Open module M2: gap estimate.** Establish a same-theory estimate implying that the reconstructed Hamiltonian has no spectrum in $(0, \Delta)$ for some $\Delta > 0$. No endpoint estimate may assume exponential decay or a gap in order to prove this step.

8 Conclusion

The finite Schur–Feshbach coefficient test has been implemented in a concrete reflection-positive lattice gauge theory rather than left as a conditional Wilson template. The construction fixes the compact simple group $SU(2)$, a four-dimensional heat-kernel plaquette action, the time reflection, three closed hairpin loops, a symmetry-adapted positive-time Wilson family, and an exact observable-level normalization. The resulting block is explicitly calculable and positive at every finite cutoff.

The main quantitative output is

$$A = 2\sigma uu^T, \qquad C = -2\sigma^2 uu^T + \frac{2\tau}{3} vv^T.$$

The first reduced margin is 2σ and the corrected null-sector margin is $2\tau/3$. The calculation also exhibits the purpose of the Schur correction: it removes the spurious σ^2 contribution from the second null-channel value.

What remains is now sharply separated. The verified calculation is the finite source-collision algebra: it certifies the displayed 3×3 block, the cancellation of the spurious σ^2 null-channel term, and the reduced margins. The present source collision must still be replaced or supplemented by a lattice-resolved geometric collar; renormalized coefficients and strict margins must be controlled along a continuum trajectory; endpoint cumulants need two-sided, regulator-specific error estimates; and reflection positivity must be joined to continuum reconstruction and a spectral estimate. Those are open modules, not consequences of the finite block or of the verification certificate.

A Normalization algebra

Lemma A.1 (One-body normalization is a congruence)**.** *Let $N = N^*$ and let $D = \mathrm{diag}(d_1, \dots, d_n)$ be invertible. Then*

$$N \succeq 0 \quad \Longleftrightarrow \quad DND^* \succeq 0.$$

Proof. For every x, $x^* DND^* x = (D^* x)^* N (D^* x)$, and D^* is bijective. □

In particular, the correlation normalization

$$q_{ij} = \frac{N_{ij}}{\sqrt{N_{ii}N_{jj}}}$$

is a congruence whenever $N_{ii} > 0$. It is realized by the normalized observables $F_i/\sqrt{N_{ii}}$.

Lemma A.2 (An uncontrolled entrywise quotient need not preserve positivity)**.** *There are positive-semidefinite matrices N and B, with every entry of B positive, such that $N \oslash B$ is indefinite.*

Proof. Take

$$N = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1/10 \\ 1/10 & 1 \end{pmatrix}.$$

Both are positive semidefinite, whereas

$$N \oslash B = \begin{pmatrix} 1 & 10 \\ 10 & 1 \end{pmatrix}$$

has eigenvalues 11 and -9. □

Lemma A.3 (Zero-sum cancellation of additive one-body terms)**.** *For $d \in \mathbb{C}^n$, let*

$$L = \frac{1}{2}(d\mathbf{1}^* + \mathbf{1}d^*).$$

Then $PLP = 0$.

Proof. For $v, w \in \mathbf{1}^\perp$, both $\mathbf{1}^* w$ and $v^*\mathbf{1}$ vanish, so $v^* L w = 0$. □

B Exact block algebra and uniform remainder

From (32),

$$N_\varepsilon = \begin{pmatrix} 1+\varepsilon\sigma & 1-\varepsilon\sigma & 1 \\ 1-\varepsilon\sigma & 1+\varepsilon\sigma & 1 \\ 1 & 1 & 1+\varepsilon^2\tau \end{pmatrix}.$$

Applying (34) gives (37). For a fully explicit uniform Taylor constant, define

$$f(\varepsilon, \sigma) = \frac{1-\varepsilon\sigma}{1+\varepsilon\sigma},$$
$$g(\varepsilon, \sigma, \tau) = \big((1+\varepsilon\sigma)(1+\varepsilon^2\tau)\big)^{-1/2}$$

and

$$M_* = \max_{\substack{0\le t\le 1/4 \\ 0\le \sigma\le 1 \\ 0\le \tau\le 4}} \max\left\{|\partial_t^3 f(t,\sigma)|, |\partial_t^3 g(t,\sigma,\tau)|\right\}. \tag{66}$$

The maximum is finite by compactness and positivity of the denominators. Taylor's theorem gives entrywise remainders at most $M_*\varepsilon^3/6$. The remainder matrix has zero diagonal and off-diagonal pattern (r_{12}, r_{13}, r_{13}), so its maximum row sum is at most $M_*\varepsilon^3/3$. Thus one may take

$$C_* = M_*/3 \tag{67}$$

in Proposition 5.1.

For the Schur coefficients, direct multiplication gives

$$PHP = \sigma bb^T,$$
$$P\left(K - \frac{1}{3}Hee^TH\right)P = -\sigma^2 bb^T + \frac{\tau}{9}zz^T.$$

Since $b^Tb = 2$, $z^Tz = 6$, and $b^Tz = 0$, the reduced eigenvalues are 2σ and $2\tau/3$.

The determinant can also be obtained before normalization. Since

$$\det N_\varepsilon = 4\varepsilon^3\sigma\tau$$

and

$$\det D_\varepsilon^2 = \frac{1}{(1+\varepsilon\sigma)^2(1+\varepsilon^2\tau)},$$

Equation (40) follows.

C Five-point fourth-derivative certificate

Lemma C.1 (Central five-point certificate)**.** *Let $f \in C^6([-2h, 2h])$ and define*

$$D_h f = \frac{f(2h) - 4f(h) + 6f(0) - 4f(-h) + f(-2h)}{h^4}. \tag{68}$$

Then

$$\left|D_h f - f^{(4)}(0)\right| \le \frac{h^2}{6}\sup_{|s|\le 2h}|f^{(6)}(s)|. \tag{69}$$

Proof. It is enough to prove the unit-mesh result and apply it to $g(u) = f(hu)$. The functional

$$L(g) = g(2) - 4g(1) + 6g(0) - 4g(-1) + g(-2) - g^{(4)}(0)$$

annihilates polynomials of degree at most five. Peano's kernel theorem gives

$$L(g) = \int_{-2}^{2} K(t)g^{(6)}(t)\,dt,$$

where the kernel is nonnegative and has integral $1/6$. Hence

$$|L(g)| \le \frac{1}{6}\sup_{|t|\le 2}|g^{(6)}(t)|.$$

Since $g^{(4)}(0) = h^4 f^{(4)}(0)$ and $g^{(6)}(u) = h^6 f^{(6)}(hu)$, division by h^4 proves (69). □

For an endpoint source logarithm $\ell_v(s) = \log Z_v(s)$, the lemma turns a formal fourth cumulant into a finite-difference estimator. A usable certificate still requires a uniform bound on $\ell_v^{(6)}$ and a two-sided estimate of all tails, as stated in Open module E2.

D Checklist for the next manuscript revision

P1. Supply affiliation, contact information, acknowledgements, and a data/code statement.

P2. Choose at least one numerical finite lattice (T, L, κ, r, s, h) and evaluate σ and τ with certified uncertainties.

P3. Decide whether the principal next theorem concerns a source collision, a geometric collar collision, or both; keep the parameters notationally distinct.

P4. For a geometric collar, provide path diagrams at every cutoff and specify the joint order of $a \downarrow 0$ and $\delta \downarrow 0$.

P5. Derive any endpoint cumulant representation from the chosen action and path incidence; do not infer a negative obstruction from failure of a one-sided positive lower bound.

P6. Add a continuum theorem only after the required uniform estimates are proved. Keep the mass-gap claim separate from finite reflection positivity.

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